University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2018, Part 2

Answer four problems. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let R be a commutative ring with 1 and let I, J, P be ideals in R such that $I \cap J \subset P$.

(a) Prove that if P is a prime ideal then either $I \subset P$ or $J \subset P$.

(b) Give an example which shows that if P is not prime then the conclusion above may not hold.

2. Prove that the ring $\mathbb{Z}[\sqrt{-2}] = \{a + b\sqrt{-2} : a, b \in \mathbb{Z}\}$ is a Euclidean domain with respect to the norm defined by $N(a + b\sqrt{-2}) = a^2 + 2b^2$ for $a, b \in \mathbb{Z}$.

3. Prove that the following polynomials are irreducible over $\mathbb{Q}$. Cite any theorems that you use.

(a) $X^3 - 5X^2 + X + 2$

(b) $X^4 + 33X^2 - 6$

(c) $X^4 + 2X^3 - 14X^2 + 31X - 7$

4. Let R be a ring with 1.

(a) State the First Isomorphism Theorem for R-modules.

(b) Let $A_1, A_2, \dots, A_n$ be R-modules and for $1 \leq i \leq n$ let B_i be a submodule of A_i. Prove that

$$(A_1 \times \cdots \times A_n)/(B_1 \times \cdots \times B_n) \cong (A_1/B_1) \times \cdots \times (A_n/B_n).$$

5. State the classification theorems for finitely generated modules over a PID. You should give both the invariant factor and the elementary divisor decompositions. Be sure to include the uniqueness statements.

6. Let L/K be an algebraic field extension and let R be a subring of L such that $K \subset R$. Prove that R is a field.